Project: Worked examples
Title: EN 1993-1-1 LTB and N+M interaction
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1. Scope and method

Lateral-torsional buckling (6.3.2), flexural buckling (6.3.1) and the interaction of axial compression with bending (6.3.3) for a uniform doubly-symmetric I-section member on fork supports, class 1 to 3: M_cr, $\bar{\lambda}_{LT}$, χ_LT and M_b,Rd; χ_y, χ_z and N_b,Rd; the Annex B (or Annex A) interaction factors and both interaction ratios, with the governing check.
Critical moment (NCCI SN003 form): $M_{cr} = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 \cdot I_w/I_z + (kL)^2 \cdot G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$; $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ (end moments, k = 1) unless entered LTB slenderness: $\bar{\lambda}_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$, $W_y = W_{pl,y}$ (class 1, 2) or $W_{el,y}$ (class 3) 6.3.2.2: $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - 0.2) + \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \bar{\lambda}_{LT}^2)}) \le 1$ 6.3.2.3: $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1, \le 1/\bar{\lambda}_{LT}^2$; optional $\chi_{LT,mod} = \chi_{LT}/f \le 1, \le 1/\bar{\lambda}_{LT}^2$ (eq. 6.58), $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$ Buckling resistance in bending: $M_{b,Rd} = \chi_{LT} \cdot W_y \cdot f_y/\gamma_{M1}$ Flexural buckling: $N_{cr} = \pi^2 EI/L_{cr}^2$, $\bar{\lambda} = \sqrt{(A \cdot f_y/N_{cr})}$, $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$, $N_{b,Rd} = \chi \cdot A \cdot f_y/\gamma_{M1}$; the curve of each axis from Table 6.2 (section type, h/b, t_f; $f_y \le 420$ MPa) or chosen Interaction (6.3.3(4)): $N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1}) + k_{yy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{yz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$; $N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1}) + k_{zy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{zz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$ Factors k_ij: Annex B from the slenderness, $N_{Ed}/(\chi \cdot N_{Rk}/\gamma_{M1})$ and C_m ($C_m = 0.6 + 0.4\psi \ge 0.4$ for a linear diagram, or entered); Annex A from the μ, w, n_{pl}, a_{LT} and C_{ij} expressions (class 1, 2: plastic column of Table A.1; class 3: elastic column, without the $0.6\sqrt{(w_i/w_j)}$ factors of k_{yz}, k_{zy})

Ref: EN 1993-1-1:2005+A1:2014 6.3.1.2, 6.3.1.3 — Buckling curves; slenderness for flexural buckling
Ref: EN 1993-1-1:2005+A1:2014 6.3.2.2 — Lateral-torsional buckling curves: general case
Ref: EN 1993-1-1:2005+A1:2014 6.3.2.3 — Lateral-torsional buckling curves for rolled sections and equivalent welded sections
Ref: EN 1993-1-1:2005+A1:2014 6.3.3 — Uniform members in bending and axial compression
Ref: EN 1993-1-1:2005+A1:2014 Annex A, Annex B — Interaction factors k_ij: methods 1 and 2
Ref: NCCI SN003 Mcr expression — Elastic critical moment of a doubly-symmetric member
Ref: EN 1993-1-1:2022 8.3 — Stability of members (label only, see the edition note)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
edition	EN 1993-1-1 edition (2005+A1:2014 until the 2022 edition is adopted; the 2022 choice changes only the label, see the sheet note)	2005+A1:2014		spec: BS EN 1993-1-1:2005+A1:2014 with the UK National Annex	any
ltb_method	χ_LT method	6.3.2.3 (rolled and welded sections)		assumption: 6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.	any
annex	Interaction factors	Annex B (method 2)		assumption: Annex B (method 2) chosen; Annex A can be selected for comparison.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
torsion	Member behaviour for Annex B	susceptible to torsional deformation (Table B.2)		assumption: An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).	any
f_y	Yield strength (for the thickness)	345	MPa	spec: BS EN 10025-2 S355, 16 mm < t ≤ 40 mm (flange 19.6 mm)	235 … 460 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 cl. 3.2.6	190 … 220 GPa
G	Shear modulus	81	GPa	spec: BS EN 1993-1-1 cl. 3.2.6: G = E/(2(1 + ν)) ≈ 81 GPa	70 … 90 GPa
gamma_M1	Partial factor γ_M1 (member buckling)	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.25
section_class	Cross-section class (class 4 needs effective properties and is outside this sheet)	1		assumption: Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).	any
section_type	Section type (selects the buckling curve of Table 6.4 / 6.5)	rolled I		spec: BS EN 10365 UB 457×191×98, hot-rolled	any
h	Overall depth h	467.2	mm	spec: BS EN 10365 UB 457×191×98	50 … 2500 mm
b	Flange width b	192.8	mm	spec: BS EN 10365 UB 457×191×98	30 … 1000 mm
t_f	Flange thickness t_f (S1: t_f; selects the flexural buckling curves of Table 6.2)	19.6	mm	spec: BS EN 10365 UB 457×191×98	2 … 150 mm
A	Area A	125.265	cm^2	reference: S1 section sheet SEC-01, UB 457×191×98 (A = 125.265 cm²)	5 … 3000 cm^2
I_y	Second moment of area, major axis I_y (S1: I_xx)	45728	cm^4	reference: S1 section sheet SEC-01 (I_xx = 45727.99 cm⁴)	50 … 5000000 cm^4
I_z	Second moment of area, minor axis I_z (S1: I_yy)	2347.02	cm^4	reference: S1 section sheet SEC-01 (I_yy = 2347.02 cm⁴)	5 … 500000 cm^4

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I_t	Torsion constant I_t (S1: J)	121.287	cm^4	reference: S1 section sheet SEC-01 (J = 121.287 cm⁴)	0.1 … 100000 cm^4
I_w	Warping constant I_w	1.17554	dm^6	reference: S1 section sheet SEC-01 (I_w = 1.17554 dm⁶)	0.00001 … 100000 dm^6
W_pl_y	Plastic modulus, major axis (S1: W_pl_x)	2232.46	cm^3	reference: S1 section sheet SEC-01 (W_pl,x = 2232.46 cm³)	1 … 100000 cm^3
W_el_y	Elastic modulus, major axis (S1: W_el_x)	1957.53	cm^3	reference: S1 section sheet SEC-01 (W_el,x = 1957.53 cm³)	1 … 100000 cm^3
W_pl_z	Plastic modulus, minor axis (S1: W_pl_y)	378.904	cm^3	reference: S1 section sheet SEC-01 (W_pl,y = 378.904 cm³)	0.5 … 50000 cm^3
W_el_z	Elastic modulus, minor axis (S1: W_el_y)	243.467	cm^3	reference: S1 section sheet SEC-01 (W_el,y = 243.467 cm³)	0.5 … 50000 cm^3
L	Length between lateral restraints (fork supports)	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
k	Effective length factor k (end rotation about the minor axis)	1		assumption: Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).	0.5 … 1
k_w	Warping factor k_w	1		assumption: Free to warp at the restraints.	0.5 … 1
z_g	Load height z_g above the shear centre, towards the compression flange (positive destabilises)	0	mm	assumption: End-moment loading: no transverse load, so the load height does not enter.	-1000 … 1000 mm
C_2	C_2 (0 for end moments; enter it for transverse loads)	0		assumption: C_2 is 0 for end-moment loading.	0 … 1.5
ltb_curve	Buckling curve for χ_LT	auto (Table 6.4 / 6.5)		assumption: Curve from Table 6.5 for a rolled I with h/b > 2.	any
use_f	Apply the f modification to χ_LT (6.3.2.3(2))	yes		assumption: The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
alpha_a0	Imperfection factor α, curve a0 (recommended 0.13)	0.13		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_a	Imperfection factor α, curve a (recommended 0.21)	0.21		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_b	Imperfection factor α, curve b (recommended 0.34)	0.34		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_c	Imperfection factor α, curve c (recommended 0.49)	0.49		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_d	Imperfection factor α, curve d (recommended 0.76)	0.76		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
lambda_LT0	$\bar{\lambda}_{LT,0}$ for 6.3.2.3 (recommended 0.4)	0.4		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.2 … 0.5
beta_LT	β for 6.3.2.3 (recommended 0.75)	0.75		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.5 … 1
moment_mode	Moment diagram given as	end moments		assumption: The moments are the member end moments.	any
N_Ed	Design axial force, TENSION positive and compression negative (as S2 N_e1 and S3; 6.3.3 applies to compression)	-400	kN	reference: S2 frame sheet FR-01, member 2, ULS combination (compression negative)	-100000 … 0 kN
M_y_A	Moment at end A about y-y, as an ordinate of the bending-moment diagram (end-moment mode): the same sign at both ends is single curvature, opposite signs double curvature. Not the end actions of a stiffness analysis, which flip sign at one end	340	kN*m	reference: S2 frame sheet FR-01, member 2, end A	-100000 … 100000 kN*m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
M_y_B	Moment at end B about y-y, same convention as A: a bending-moment-diagram ordinate, the same sign as A for single curvature (end-moment mode)	170	kN*m	reference: S2 frame sheet FR-01, member 2, end B (same side: single curvature)	-100000 ... 100000 kN*m
M_z_Ed	Greatest design moment about z-z	6	kN*m	assumption: Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).	0 ... 100000 kN*m
psi_z	ψ of the M_z diagram	1		assumption: Uniform minor-axis moment (conservative).	-1 ... 1
L_cr_y	Buckling length about y-y	5	m	drawing: FR-BM-12 rev B	0.1 ... 100 m
L_cr_z	Buckling length about z-z	5	m	drawing: FR-BM-12 rev B	0.1 ... 100 m
curve_y	Buckling curve about y-y (auto: Table 6.2 from the section type, h/b and t_f, for f_y up to 420 MPa)	auto (Table 6.2)		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about y-y gives curve a	any
curve_z	Buckling curve about z-z (auto: Table 6.2 from the section type, h/b and t_f, for f_y up to 420 MPa)	auto (Table 6.2)		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about z-z gives curve b	any

3. Assumptions

A1. Uniform doubly-symmetric I-section member on fork supports (free to warp and to rotate about the minor axis at the ends, restrained against twist); no mono-symmetric or tapered members, no cantilevers, no continuous-restraint effects. (Scope of S7; k and k_w below 1 model partial end restraint of the minor-axis rotation and of warping.)
A2. Class 1, 2 or 3 as stated by the user; class 4 effective properties are outside this sheet. (6.2.2.5 is not implemented.)
A3. C_1 from the ψ formula applies to end-moment diagrams with k = 1; for transverse loads, k ≠ 1 or other diagrams the user enters C_1, C_2 (and k_c, C_m) from the source in use. With C_1 entered the sheet blocks until k_c (when the f modification is used) and C_my (Annex B) are entered too, unless the diagram is given as end moments: ψ is then a real, entered moment shape. (No table of C_1 or C_2 values is shipped (reference records only).)
A4. The imperfection factors, β and $\bar{\lambda}_{LT,0}$ default to the recommended values and are a user-verifiable factor set; the National Annex in use may differ. (Record kind ec3.ltb-factors.)
A5. The 2022 edition choice changes the label printed on the sheet only, by decision: the arithmetic implemented is that of EN 1993-1-1:2005+A1:2014, the differences between the editions are not implemented (they cannot be stated with certainty), and the clause numbers in the 2022 edition differ and must be checked by the user. (The sheet says so (warning `edition`).)

A6. The flexural buckling curves selected automatically are those of Table 6.2 for S 235 to S 420 (rolled I: a, b for h/b > 1.2 and t_f ≤ 40 mm, b, c above 40 mm or h/b ≤ 1.2 up to 100 mm, d, d beyond; welded I: b, c up to 40 mm, c, d above); for f_y above 420 MPa or a combination the table does not cover the curves are chosen by hand. (Table 6.2 is a small selection rule and is not reproduced beyond it.)
A7. The Annex A and Annex B expressions are implemented as published; their coefficients are to be checked against the owner's copy of the standard. (Interaction factors for diagrams other than linear are entered, not derived.)
A8. ltb_method (χ_LT method) is taken as 6.3.2.3 (rolled and welded sections). (6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.)
A9. annex (Interaction factors) is taken as Annex B (method 2). (Annex B (method 2) chosen; Annex A can be selected for comparison.)
A10. torsion (Member behaviour for Annex B) is taken as susceptible to torsional deformation (Table B.2). (An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).)
A11. section_class (Cross-section class (class 4 needs effective properties and is outside this sheet)) is taken as 1. (Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).)
A12. k (Effective length factor k (end rotation about the minor axis)) is taken as 1. (Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).)
A13. k_w (Warping factor k_w) is taken as 1. (Free to warp at the restraints.)
A14. z_g (Load height z_g above the shear centre, towards the compression flange (positive destabilises)) is taken as 0 mm. (End-moment loading: no transverse load, so the load height does not enter.)
A15. C_2 (C_2 (0 for end moments; enter it for transverse loads)) is taken as 0. (C_2 is 0 for end-moment loading.)
A16. ltb_curve (Buckling curve for χ_LT) is taken as auto (Table 6.4 / 6.5). (Curve from Table 6.5 for a rolled I with h/b > 2.)
A17. use_f (Apply the f modification to χ_LT (6.3.2.3(2))) is taken as yes. (The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.)
A18. moment_mode (Moment diagram given as) is taken as end moments. (The moments are the member end moments.)
A19. M_z_Ed (Greatest design moment about z-z) is taken as 6 kN*m. (Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).)
A20. psi_z (ψ of the M_z diagram) is taken as 1. (Uniform minor-axis moment (conservative).)
A21. Flexural buckling curves a (y-y) and b (z-z) from Table 6.2: rolled I-section, h/b = 2.4232, t_f = 19.6 mm. (Automatic selection; choose the curves by hand to override.)

4. Working

[36] $M_{y,Ed} = \max(|M_A|, |M_B|)$ — 6.3.2
$= \max(|340|, |170|)\ kN·m$
$= 340\ kN·m$
Greatest design moment about y-y

[37] $\psi_y = \square = M_{small}/M_{large}$ (same sign: single curvature) — Table B.3
$= 170 / 340$
$= 0.5$
Ratio of the end moments

[38] $C_1 = 1.88 - 1.40\square + 0.52\square^2 \le 2.70$ — NCCI SN003 / ENV 1993-1-1 Annex F
$= \square = 0.5$
$= 1.31$
C_1 for end moments (derived)

[39] $M_{cr} = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 I_w/I_z + (kL)^2 G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$ — NCCI SN003
$= 1.31 \times \pi^2 \times 210000 \times 23470000 / (5000)^2 \times [\sqrt{((1/1)^2 \times 1175500000000/23470000 + (5000)^2 \times 81}$
$1212900 / (\pi^2 \times 210000 \times 23470000) + (0 \times 0)^2) - 0]$
$= 808.38\ kN·m$

Elastic critical moment for lateral-torsional buckling

[40] W_y = W_pl,y (class 1, 2) 6.3.2.2(1)

= plastic

= 2232.5 cm³

Section modulus used for bending about y

[41] lambda_LT = √(W_y·f_y/M_cr) 6.3.2.2(1)

= √(2232.5 cm³ × 345 MPa / 808.38 kN·m)

= 0.9761

Non-dimensional slenderness for lateral-torsional buckling

[48] alpha_LT = imperfection factor, curve c 6.3.2.3(1), Table 6.5

= Table 6.5, rolled, h/b = 2.4232

= 0.49

Imperfection factor for lateral-torsional buckling

[51] Phi_LT = 0.5[1 + □_LT(□̄_LT - □̄_LT,0) + □·□̄_LT²] 6.3.2.3(1)

= 0.5 × [1 + 0.49 × (0.9761 - 0.4) + 0.75 × 0.9761²]

= 0.99843

Φ_LT

[52] chi_LT_0 = 1/(□_LT + √(□_LT² - □·□̄_LT²)) ≤ 1, ≤ 1/□̄_LT² 6.3.2.3(1)

= 1/(0.99843 + √(0.99843² - 0.75 × 0.9761²))

= 0.6537

Lateral-torsional reduction factor

[54] k_c = 1/(1.33 - 0.33□) Table 6.6

= □ = 0.5

= 0.85837

k_c for a linear moment diagram (derived)

[55] f = 1 - 0.5(1 - k_c)[1 - 2(□̄_LT - 0.8)²] ≤ 1 6.3.2.3(2)

= 1 - 0.5 × (1 - 0.85837) × [1 - 2 × (0.9761 - 0.8)²]

= 0.93358

Modification factor f

[56] chi_LT_mod = □_LT/f ≤ 1, ≤ 1/□̄_LT² (eq. 6.58) 6.3.2.3(2)

= 0.6537 / 0.93358, 1/□̄_LT² = 1.0496

= 0.70022

Modified lateral-torsional reduction factor

[57] chi_LT = □_LT,mod 6.3.2.3

= as above

= 0.70022

Lateral-torsional reduction factor used

[58] Mb_Rd = □_LT·W_y·f_y/□_M1 6.3.2.1(3)

= 0.70022 × 2232.5 cm³ × 345 MPa / 1

= 539.3 kN·m

Design buckling resistance moment

[59] u_LTB = M_y_Ed / Mb_Rd

= 340 kN·m / 539.3 kN·m

= 0.63044

Lateral-torsional buckling utilisation

[62] alpha_y = imperfection factor, curve a 6.3.1.2, Table 6.2

= Table 6.2

= 0.21

Imperfection factor about y

[63] alpha_z = imperfection factor, curve b 6.3.1.2, Table 6.2

= Table 6.2

= 0.34

Imperfection factor about z

[64] $N_{Rk} = A \cdot f_y$ 6.3.1.1(3)
$= 125.27\ cm^2 \times 345\ MPa$
$= 4321.6\ kN$
Characteristic axial resistance
[65] $N_{cr,y} = \pi^2 E \cdot I_y / L_{cr,y}^2$ 6.3.1.2
$= \pi^2 \times 210000 \times 457280000 / 5000^2$
$= 37911\ kN$
Elastic critical force about y
[66] $N_{cr,z} = \pi^2 E \cdot I_z / L_{cr,z}^2$ 6.3.1.2
$= \pi^2 \times 210000 \times 23470000 / 5000^2$
$= 1945.8\ kN$
Elastic critical force about z
[67] $\lambda_y = \sqrt{(A \cdot f_y / N_{cr,y})}$ 6.3.1.3
$= \sqrt{(4321.6 / 37911)}$
$= 0.33763$
Non-dimensional slenderness about y
[68] $\lambda_z = \sqrt{(A \cdot f_y / N_{cr,z})}$ 6.3.1.3
$= \sqrt{(4321.6 / 1945.8)}$
$= 1.4903$
Non-dimensional slenderness about z
[69] $\chi_y = 1 / (\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$ 6.3.1.2
$= \Phi = 0.21,\ \bar{\lambda} = 0.33763$
$= 0.96853$
Reduction factor about y
[70] $\chi_z = 1 / (\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$ 6.3.1.2
$= \Phi = 0.34,\ \bar{\lambda} = 1.4903$
$= 0.34582$
Reduction factor about z
[71] $N_{b,y,Rd} = \chi_y \cdot A \cdot f_y / \gamma_{M1}$ 6.3.1.1(3)
$= 0.96853 \times 4321.6 / 1$
$= 4185.7\ kN$
Buckling resistance about y
[72] $N_{b,z,Rd} = \chi_z \cdot A \cdot f_y / \gamma_{M1}$ 6.3.1.1(3)
$= 0.34582 \times 4321.6 / 1$
$= 1494.5\ kN$
Buckling resistance about z
[73] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$
$= \min(4185.7\ kN, 1494.5\ kN)$
$= 1494.5\ kN$
Flexural buckling resistance (smaller of the two axes)
[74] $M_{y,Rk} = W_y \cdot f_y$ 6.3.3
$= 2232.5\ cm^3 \times 345\ MPa$
$= 770.2\ kN \cdot m$
Characteristic bending resistance about y
[75] $M_{z,Rk} = W_z \cdot f_y$ ($W_{pl,z}$ class 1, 2; $W_{el,z}$ class 3) 6.3.3
$= 378.9\ cm^3 \times 345\ MPa$
$= 130.72\ kN \cdot m$
Characteristic bending resistance about z
[76] $C_{my} = 0.6 + 0.4 \Psi_y \ge 0.4$ (or entered) Table B.3
$= \Psi_y = 0.5$
$= 0.8$
Equivalent uniform moment factor, y

[77] C_mz = 0.6 + 0.4χ_z ≥ 0.4 (or entered) Table B.3
= χ_z = 1
= 1
Equivalent uniform moment factor, z
[78] C_mLT = as C_my (or entered) Table B.3
= C_my
= 0.8
Equivalent uniform moment factor for lateral-torsional buckling
[79] k_yy = Annex B: C_my(1 + (λ̄_y - 0.2)n_y) ≤ C_my(1 + 0.8n_y) (class 1, 2) Annex B Table B.2
= n_y = 0.095565
= 0.81052
Interaction factor k_yy
[80] k_yz = Annex B: 0.6k_zz (class 1, 2), k_zz (class 3) Annex B Table B.2
= as above
= 0.82482
Interaction factor k_yz
[81] k_zy = Annex B: Table B.1 0.6k_yy / 0.8k_yy; Table B.2 from λ̄_z, n_z and C_mLT Annex B Table B.2
= n_z = 0.26764
= 0.95134
Interaction factor k_zy
[82] k_zz = Annex B: C_mz(1 + (2λ̄_z - 0.6)n_z) ≤ C_mz(1 + 1.4n_z) (class 1, 2) Annex B Table B.2
= as above
= 1.3747
Interaction factor k_zz
[83] u_N_y = N_Ed/(χ_y·N_Rk/γ_M1) 6.3.3(4)
= 400 kN / (0.96853 × 4321.6 kN / 1)
= 0.095565
Axial term about y
[84] u_N_z = N_Ed/(χ_z·N_Rk/γ_M1) 6.3.3(4)
= 400 kN / (0.34582 × 4321.6 kN / 1)
= 0.26764
Axial term about z
[85] u_M_y = M_y,Ed/(χ_LT·M_y,Rk/γ_M1) 6.3.3(4)
= 340 kN·m / (0.70022 × 770.2 kN·m / 1)
= 0.63044
Major-axis bending term
[86] u_M_z = M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 6 kN·m / (130.72 kN·m / 1)
= 0.045899
Minor-axis bending term
[87] u_61 = N_Ed/(χ_y N_Rk/γ_M1) + k_yy·M_y,Ed/(χ_LT M_y,Rk/γ_M1) + k_yz·M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 0.095565 + 0.81052 × 0.63044 + 0.82482 × 0.045899
= 0.64441
Interaction, equation 6.61
[88] u_62 = N_Ed/(χ_z N_Rk/γ_M1) + k_zy·M_y,Ed/(χ_LT M_y,Rk/γ_M1) + k_zz·M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 0.26764 + 0.95134 × 0.63044 + 1.3747 × 0.045899
= 0.9305
Interaction, equation 6.62
[89] u_interaction = max(u_61, u_62) 6.3.3(4)
= max(0.64441, 0.9305)
= 0.9305
Greatest interaction ratio

[90] u_buckling = N_Ed/N_b,Rd 6.3.1.1
= 400 kN / 1494.5 kN
= 0.26764
Flexural buckling utilisation
[91] u_member = max(max(u_interaction, u_LTB), u_buckling)
= max(max(0.9305, 0.63044), 0.26764)
= 0.9305
Governing utilisation of the member

5. Results

SYMBOL	VALUE	DESCRIPTION
edition_used	"EN 1993-1-1:2005+A1:2014"	EN 1993-1-1 edition
psi_y	0.5	ψ of the M_y diagram
M_y_Ed	340 kN·m	Greatest design moment about y-y
C_1	1.31	C_1
Mcr	808.38 kN·m	M_cr
lambda_LT	0.9761	λ̄_LT
chi_LT	0.70022	χ_LT
Mb_Rd	539.3 kN·m	M_b,Rd
N_Rk	4321.6 kN	N_Rk
N_cr_y	37911 kN	N_cr,y
N_cr_z	1945.8 kN	N_cr,z
lambda_y	0.33763	λ̄_y
lambda_z	1.4903	λ̄_z
chi_y	0.96853	χ_y
chi_z	0.34582	χ_z
N_b_y_Rd	4185.7 kN	N_b,y,Rd
N_b_z_Rd	1494.5 kN	N_b,z,Rd
N_b_Rd	1494.5 kN	N_b,Rd
k_yy	0.81052	k_yy
k_yz	0.82482	k_yz
k_zy	0.95134	k_zy
k_zz	1.3747	k_zz
u_LTB	0.63044	LTB utilisation
u_buckling	0.26764	Buckling utilisation
u_61	0.64441	Equation 6.61
u_62	0.9305	Equation 6.62
u_interaction	0.9305	Interaction utilisation

SYMBOL	VALUE	DESCRIPTION
u_member	0.9305	Member utilisation
alpha_LT	0.49	α_LT
alpha_y	0.21	α_y
alpha_z	0.34	α_z
Phi_LT	0.99843	Φ_LT
chi_LT_0	0.6537	χ_LT before the f modification
k_c	0.85837	k_c
f	0.93358	f
W_y	2232.5 cm³	W_y
M_y_Rk	770.2 kN·m	M_y,Rk
M_z_Rk	130.72 kN·m	M_z,Rk
C_my	0.8	C_my
C_mz	1	C_mz
C_mLT	0.8	C_mLT
u_N_y	0.095565	Axial term, y
u_N_z	0.26764	Axial term, z
u_M_y	0.63044	Bending term, y
u_M_z	0.045899	Bending term, z
governing	"interaction 6.62 (0.931)"	Governing check

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Interaction, equation 6.62	u_62 = 0.9305	u_62 ≤ 1	0.931	6.3.3(4)
PASS	Interaction, equation 6.61	u_61 = 0.64441	u_61 ≤ 1	0.644	6.3.3(4)
PASS	Lateral-torsional buckling M_y,Ed ≤ M_b,Rd	u_LTB = 0.63044	u_LTB ≤ 1	0.63	6.3.2.1
PASS	Flexural buckling N_Ed ≤ N_b,Rd	u_buckling = 0.26764	u_buckling ≤ 1	0.268	6.3.1.1

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, G, f_y (Material properties); gamma_M1 (EN 1993-1-1 National Annex factors); alpha_a0, alpha_a, alpha_b, alpha_c, alpha_d, lambda_LT0, beta_LT (Imperfection factors and 6.3.2.3 parameters); C_2 (C_1, C_2 and k_c for the moment diagram). Enter each as one so its source, edition and licence travel with it.

8. Verification

```
Template struct.ltb-interaction@1.1.0 Content hash feab0d5f384d5051 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```